Abed Garra

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Education

University of Massachusetts Amherst Sep 2026 – Present **(Expected 05/31)**
Ph.D. in Computer Science; Advisor: Prof. Eugene Bagdasarian *Amherst, MA, USA*

Tel Aviv University 2024 – 2026
M.Sc. in Computer Science, GPA 96/100; Advisor: Dr. Mahmood Sharif *Tel Aviv, Israel*
Thesis: ***The TellTail of Embeddings: Fingerprinting Retrievers in Black-Box Systems***
B.Sc. in Computer Science

Research Experience

Graduate Researcher, PLUS Lab, Tel Aviv University Dec 2023 – 2026

Fingerprinting retrievers in black-box systems (M.Sc. thesis) [1]

- Built TellTail, a black-box **fingerprinting** attack that identifies the embedding model behind retrieval and RAG systems.
- Evaluated on **53 embedding retrievers** in an **open-world** setting (19 known, 34 unknown).
- Achieved **92.5%** attack success rate (ASR) from LLM responses alone.
- Fingerprinted the retriever behind **OpenWebUI**, an open-source RAG deployment.

Forecasting cyber incidents with LLMs [2]

- Engineered an LLM pipeline turning news and podcasts into features to forecast **15,700+** security incidents.
- Ran **Internet-wide scans** on multiple ports with ZMap and built DNS infrastructure for network measurements.
- Reached **81.3%** ROC AUC vs. 77.5% for the prior state of the art; trained Longformer with multi-GPU DDP.

Open-Source Datasets

Dense retrieval indexes (Hugging Face): Pre-built FAISS indexes of MS MARCO for 53 embedding models [1].
News corpus (Hugging Face): Deduplicated extension of CC-News from June 2024 onward, **actively maintained** [2].

Publications (* equal contribution)

[1] **A. Garra**, M. Ben-Tov, M. Sharif. The TellTail of Embeddings: Fingerprinting Retrievers in Black-Box Systems. *arXiv*, 2026 (submitted to SaTML). [paper] [code]

[2] **A. Garra***, M. Fesenko*, Y. Harel, M. Sharif. Forecasting Cybersecurity Incidents Using Geopolitical Data and Large Language Models. *arXiv*, 2026. [paper]

[3] Y. Wang, H. Liu, X. Xu, et al. (43 authors, incl. **A. Garra**). Towards Trustworthy Physical Intelligence: From Theory to Practice Across Life Cycle. *arXiv*, 2026. [paper]

[4] H. Abu-Kishk, M. Dahan, **A. Garra**. Between Reflection and Construction: AI as the New Orientalism? *Journal of Information, Communication and Ethics in Society*, 2026. [paper]

[5] **A. Garra**. Mitigating Emergent Collusion in LLM Pricing Agents. *arXiv*, 2026. [paper]

Teaching

Teaching Assistant, Tel Aviv University Feb 2025 – Sep 2025

- Mentored student teams through real-world ML-for-security projects in two graduate workshops: *Usable Security & Privacy* (Dr. Mahmood Sharif) and *Intrusion Detection with ML* (Dr. David Movshovitz).

Software Project Grader (Apr – Oct 2024) and **Calculus Tutor** (Sep 2022 – Apr 2023), Tel Aviv University

Languages

Arabic (native), English (fluent; TOEFL iBT 111/120), Hebrew (fluent)